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Hydraulic damper with a hydromechanical compression stop assembly

Abstract

A hydraulic damper comprises a main tube, a piston assembly, a base valve assembly, and a compression stop assembly. The compression stop assembly includes an insert defining an inner chamber, and a sleeve displaceable along with the main piston assembly and configured to be slidably introduced inside the inner chamber. The sleeve has a diameter lower than the diameter of the main tube defining a first external flow channel between the sleeve and the main tube; the sleeve is attached to the piston assembly by a spring disposed within the sleeve; and the insert is provided with a plurality of axially-spaced holes and has an annular flange adjoining the inner wall of the main tube and separating the compression chamber from a second external flow channel between the radially external outlets of the holes and the base valve assembly.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to Chinese Patent Application No. 202111176150.4, filed on Oct. 9, 2021, which is hereby incorporated by reference in its entirety.

FIELD OF THE INVENTION

(2) The present disclosure relates to a hydraulic damper, in particular a motor vehicle suspension damper.

BACKGROUND OF THE INVENTION

(3) Document US2001025753 discloses a damper assembly comprising a cylinder in which a piston rod is mounted in an axially adjustable fashion. At that end of the piston rod which is arranged in the cylinder, a piston is provided which separates a compression chamber from a rebound chamber. A rebound stop assembly which includes a rebound spring is disposed in the rebound chamber. At the end distal to the piston, the rebound spring includes an auxiliary piston, the external diameter of which is smaller than an internal diameter of the cylinder. In an end section of the rebound chamber, an auxiliary cylinder is formed in the interior of the cylinder, the internal diameter of the auxiliary cylinder matches the external diameter of the auxiliary piston so that the auxiliary piston interacts with the cylinder wall of the auxiliary cylinder in a radially sealing

fashion. Given a sufficiently large rebound stroke, the auxiliary piston moves into the auxiliary cylinder and thus closes off the interior of the auxiliary cylinder on the side facing the piston.

(4) Documents US2018355944, DE102015119731 and DE1430494 disclose damper assemblies provided with various end of stroke stop assemblies comprising compressible spring arrangements coupled with pistons.

(5) As a compression stop assembly requires space for operation, it is common to provide this space by decreasing a so-called minimum bearing span of a damper, that is the distance between a rebound stop and a main piston assembly. This in turn restricts implementation of the damper in suspension systems with a piston rod that is subjected to side loads (e.g. MacPherson struts), where a sufficient bearing span is crucial for proper operation of the damper. It is thus desirable to reduce the working length of a compression stop assembly. Reduced working length of a compression stop assembly is also beneficial in terms of packaging and handling the dampers.

(6) It has been the object of the present disclosure to provide a hydraulic damper with a compression stop assembly, which would feature reduced operational length, would be cost efficient and simple in manufacture and assembly, and which would provide versatile tuning properties for shaping the additional damping force.

SUMMARY OF THE INVENTION

(7) The present invention provides a hydraulic damper assembly. The hydraulic damper comprises a main tube filled with working liquid and extending between an open end and a closed end. The hydraulic damper also comprises a piston assembly slidably disposed inside the main tube. The piston assembly is attached to a piston rod that extends outside the hydraulic damper through a sealed piston rod guide located at the open end, dividing the main tube into a rebound chamber and a compression chamber and configured to generate a damping force. The hydraulic damper also comprises a base valve assembly located at the closed end of the compression chamber and configured to control a flow of the working liquid between the compression chamber and a compensation chamber. The hydraulic damper also comprises a compression stop assembly located in the compression chamber. The compression stop assembly includes an insert defining an inner chamber. The compression stop assembly also includes a sleeve that is displaceable along with the piston assembly and configured to be slidably introduced inside the inner chamber of the insert at an end of a compression stroke of the hydraulic damper to generate an additional damping force. The sleeve has an external diameter that is smaller than an internal diameter of the main tube, defining a first external flow channel between the sleeve and the main tube. The insert defines a second external flow channel extending annularly thereabout between the sleeve and the main tube. The insert defines a plurality of axially-spaced holes each providing fluid communication between the inner chamber and the base valve assembly via the second external flow channel.

(8) In some embodiments, the piston assembly includes compression and rebound valve assemblies to control the flow of working liquid passing between the rebound chamber and the compression chamber to generate the damping force.

(9) In some embodiments, the sleeve is attached to the piston assembly by a spring disposed within the sleeve.

(10) In some embodiments, the spring is a coil spring.

(11) In other embodiments, a wave spring, for example, might be employed.

(12) In some embodiments, the insert has an annular flange adjoining an inner wall of the main tube and separating the compression chamber from the second external flow channel.

(13) In some embodiments, the sleeve has a hollow interior that defines a radially internal flow channel.

(14) In some embodiments, the sleeve comprises a plurality of radial projections making a sliding fit with the main tube and separated by bridges.

(15) In some embodiments, the base valve assembly includes a compression valve assembly and a rebound valve assembly each configured to control the flow of the working liquid between the

compression chamber and the compensation chamber.

(16) In some embodiments, the compression stop assembly further comprises a compression safety valve that opens at a predefined threshold of pressure in the inner chamber of said insert and allows for a flow of working liquid from said inner chamber of said insert to the compression valve assembly of the base valve assembly that bypasses said second external flow channel.

(17) In some embodiments, the compression stop assembly further comprises a rebound check valve that allows a flow of the working liquid from the rebound valve assembly of the base valve assembly to said inner chamber of said insert, bypassing said second external flow channel during a rebound stroke, and blocking the flow of the working liquid therethrough during the compression stroke.

(18) In some embodiments, the compression stop assembly further comprises: a compression safety valve that opens at a predefined threshold of pressure in the inner chamber of said insert and allows for a flow of working liquid from said inner chamber of said insert to the compression valve assembly of the base valve assembly that bypasses said second external flow channel; a rebound check valve that allows a flow of the working liquid from the rebound valve assembly of the base valve assembly to said inner chamber of said insert, bypassing said second external flow channel during a rebound stroke, and blocking the flow of the working liquid therethrough during the compression stroke; and a fixing member fixed to the base valve assembly and to said insert and comprising a plurality of angularly disposed dual stroke flow passages joining the base valve assembly with said second external flow channel, wherein at least one of said compression safety valve and said rebound check valve comprises a plurality of deflective discs covering a plurality of angularly disposed flow passages in said fixing member.

(19) In some embodiments, the insert includes an internal surface defining a plurality of axial projections, and the plurality of axially-spaced holes each include a radially-internal outlet located in a corresponding one of the axial projections.

(20) In some embodiments, the insert adjoins an inner wall of the main tube and the second external flow channel is defined by a plurality of grooves made in a radially external surface of the insert and in fluid connection with the axially-spaced holes.

(21) In some embodiments, the main tube is provided with a widened section to receive the insert.

(22) In some embodiments, the insert and/or the sleeve is/are made of plastic, such as polyamide.

(23) In some embodiments, an insert is made as a single element with a fixing member and provided with a compression safety valve and/or a rebound check valve.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The invention shall be described and explained below in connection with the attached drawings on which:

(2) FIG. 1 illustrates a fragment of a vehicle suspension comprising the damper according to the present disclosure;

(3) FIG. 2 is a schematic cross-sectional view of an embodiment of a twin-tube damper according to the present disclosure with an embodiment of the compression stop assembly in a fully compressed state;

(4) FIG. 3 is a schematic cross-sectional view of the embodiment of the compression stop assembly shown in FIG. 2 in an inactive state during the compression stroke;

(5) FIG. 4 is a schematic cross-sectional view of the embodiment of the compression stop assembly shown in FIG. 2 after activation;

(6) FIG. 5 is a schematic cross-sectional view of the embodiment of the compression stop assembly shown in FIG. 2 after activation, and with a safety valve engaged;

- (7) FIG. 6 is a schematic cross-sectional view of the embodiment of the compression stop assembly shown in FIG. 2 in a fully compressed state and at the onset of the rebound stroke;
- (8) FIG. 7 is a schematic cross-sectional view of the embodiment of the compression stop assembly shown in FIG. 2 at an initial stage of the rebound stroke;
- (9) FIG. 8 is a schematic cross-sectional view of the embodiment of the compression stop assembly shown in FIG. 2 at a further stage of the rebound stroke;
- (10) FIG. 9 shows a sleeve of the compression stop assembly shown in FIG. 2 in an axonometric view;
- (11) FIG. 10 shows a sleeve of the compression stop assembly shown in FIG. 2 in a cross-sectional view;
- (12) FIG. 11 shows an insert of the compression stop assembly shown in FIG. 2 in an axonometric view;
- (13) FIG. 12 shows an insert of the compression stop assembly shown in FIG. 2 in a cross-sectional view;
- (14) FIG. 13 shows a fixing member of the compression stop assembly shown in FIG. 2 in an axonometric view from an insert side; and
- (15) FIG. 14 shows a fixing member of the compression stop assembly shown in FIG. 2 in a cross-sectional view.

DETAILED DESCRIPTION

- (16) FIG. 1 schematically illustrates a fragment of an exemplary vehicle suspension comprising a hydraulic damper 1 of the present disclosure attached to a vehicle chassis 101 by a top mount 102 and a plurality of screws 103 disposed on the periphery of the upper surface of the top mount 102. The top mount 102 is connected to a coil spring 104 and a piston rod 5 of the hydraulic damper 1. The hydraulic damper 1 includes an external tube 2 connected to the steering knuckle 105 supporting the vehicle wheel 106.
- (17) FIGS. 2-6 present an embodiment of the hydraulic damper 1 configured as a twin-tube damper, according to the present disclosure. The hydraulic damper 1 comprises an external tube 2 and a main tube 3 disposed within the external tube 2 and extending between an open end and a closed end. The main tube 3 is filled with viscous working liquid inside of which a piston assembly 4 is disposed. The piston assembly 4 is attached to a piston rod 5 that extends outside the hydraulic damper 1 through a sealed piston rod guide 6 located at the open end. The hydraulic damper 1 is also provided with a base valve assembly 7 fixed at the closed end of the main tube 3.
- (18) The piston assembly 4 makes a sliding fit with the inner surface of the main tube 3 and divides the main tube 3 into a rebound chamber 11 (between the piston assembly 4 and the piston rod guide 6) and a compression chamber 12 (between the piston assembly 4 and the base valve assembly 7). The base valve assembly 7 is configured to control a flow of the working liquid between the compression chamber 12 and an compensation chamber 13. In some embodiments, and as shown in FIG. 3, the compensation chamber 13 extends annularly between the main tube 3 and the external tube 2.
- (19) A metal retaining member 51 is clenched on a piston rod 5 and supports a rebound stop 52 of elastomeric material. The distance between the rebound stop 52 and the piston assembly 4 defines a minimum bearing span 32 of the hydraulic damper 1.
- (20) The term “compression” as used herein with reference to particular elements of the damper refers to these elements or parts of elements that are adjacent to or face the compression chamber 12 or, in a case of working liquid flow direction, it refers to this flow direction that takes place during the compression stroke of the damper. Similarly, the term “rebound” as used in this specification with reference to particular elements of the damper refers to these elements or these parts of particular elements that are adjacent to or face the rebound chamber 11 or, in a case of working liquid flow direction, it refers to this flow direction that takes place during the rebound stroke of the damper.

(21) The piston assembly **4** includes a first rebound valve assembly **41** and a first compression valve assembly **42**. Each of the first compression and first rebound valve assemblies **41**, **42** are configured to control the flow of working liquid passing between the rebound chamber **11** and the compression chamber **12** while the piston assembly **4** is in motion along an axis A and to generate a damping force opposing force applied to the piston rod **5** in corresponding compression and rebound directions. Also, the base valve assembly **7** includes a second rebound valve assembly **71** and a second compression valve assembly **72** to control the flow of working liquid passing between the compensation chamber **13** and the compression chamber **12**, respectively, during rebound and compression stroke of the hydraulic damper **1**. As it is well known to those skilled in the art, the valve assemblies **41**, **42** and **71**, **72** provide design parameters that may be used to shape desired characteristics of the hydraulic damper **1**.

(22) The hydraulic damper **1** is further provided with a compression stop assembly **8** located in the compression chamber **12** to generate an additional damping force at the end of the compression stroke e.g. in order to avoid abrupt stop of the piston assembly **4**. Main components of the compression stop assembly **8** include an insert **81**, which may have a generally cylindrical shape, and a sleeve **83** displaceable along with the piston assembly **4** and connected to the piston assembly **4** by a coil spring **84**. Other types of springs, such as a wave spring, may be used in place of the coil spring **84**.

(23) The sleeve **83**, shown in detail in FIGS. **9** and **10**, is made of polyamide and defines a radially external surface having an external diameter that is smaller than an internal diameter of the main tube **3**, defining a first external flow channel **85** between the sleeve **83** and the main tube **3**. The external diameter of the sleeve **83** may correspond to an internal diameter of the radially internal surface of the insert **81**. The sleeve **83** is slidably disposed within the main tube **3** by a plurality of radial projections **831** disposed at regular angular intervals and making a sliding fit with the main tube **3** and separated by bridges **832** that allow a flow of working liquid between the radially external surface of the sleeve **83** and the main tube **3** through the first external flow channel **85** between the compression chamber **12** and the compensation chamber **13**. Another, radially internal flow channel **86** between the compression chamber **12** and the compensation chamber **13** is provided inside the sleeve **83** between the coils of the coil spring **84** and through the tubular projection **833**. Technological holes **834** facilitate injection molding of the tubular projection **833**.

(24) The insert **81**, shown in detail in FIGS. **11** and **12**, is in this embodiment made of polyamide and fixed to the base valve assembly **7** by a fixing member **82** press-fitted to the base valve assembly **7** and to the main tube **3**. The insert **81** is also press-fitted to the fixing member **82** and in this embodiment is disposed within a widened section **31** of the main tube **3**. The insert **81** has a radially internal surface that, along with the fixing member **82**, defines an inner chamber **811** of the compression stop assembly **8**. A second external flow channel **89** extends annularly about the insert **81**, between the insert **81** and an interior surface of the main tube **3**.

(25) The insert **81** defines a plurality of axially-spaced holes **814** each providing fluid communication between the inner chamber **811** and the base valve assembly **7** via the second external flow channel **89**. In some embodiments, the axially-spaced holes **814** may be arranged in four sets at regular angular spacing (i.e. spaced apart by 90-degrees between sets). In some embodiments, each set of the axially-spaced holes **814** may include eleven of the axially-spaced holes **814**. However, the insert **81** may include a different number of sets, and each set may include a different number of the axially-spaced holes **814**.

(26) In some embodiments, the insert **81** adjoins the inner wall of the widened section **31** of the main tube **3** and in order to allow flow of liquid between the insert **81** and the main tube **3** through the base valve assembly **7**. Radially internal outlets of the axially-spaced holes **814** are located in four axial projections **816** made in a radially internal surface of the insert **81**. Radially external outlets of the axially-spaced holes **814** of each set are located in four axial grooves **813** made in a radially external surface of the insert **81**. The axial grooves **813** are terminated at the rebound axial

end of the insert **81** with an annular flange **812**. At the compression axial end of the insert **81** the axial grooves **813** are joined with an annular channel **815** between the insert **81** and the main tube **3**. The annular flange **812** may separate the second external flow channel **89** from the compression chamber **12**.

(27) The insert **81** may allow for a substantially unrestricted flow of working liquid between the compression chamber **12** and the compensation chamber **13** through the axially-spaced holes **814**, the axial grooves **813**, annular channel **815**, eight radially external, equiangularly spaced dual stroke flow passages **822** made in the fixing member **82** and the second rebound valve assembly **71** or the second compression valve assembly **72** of the base valve assembly **7**. In this embodiment, the second external flow channel **89** is defined also by the axial grooves **813** of the insert **81** and the dual stroke flow passages **822** of the fixing member **82**.

(28) The fixing member **82**, shown in detail in FIGS. **13** and **14**, is made of sintered steel and comprises a compression safety valve **87** and a rebound check valve **88**. The compression safety valve **87** comprises six deflective discs **873** covering a plurality of radially proximal flow passages **872** disposed at regular angular intervals and at a side of the fixing member **82** facing the base valve assembly **7**. The rebound check valve **88** comprises a deflective disc **881** covering a plurality of radially distal flow passages **882**, which are disposed at regular angular intervals. Deflective discs **873** and **881** are held in place by a bolt **823** passing through a retainer **824**, deflective discs **873** of the compression safety valve **87**, the fixing member **82**, the deflective disc **881** of the rebound check valve **88** and riveted over a lock washer **821**. To allow a flow of working liquid through radially proximal, radially proximal flow passages **872**, the deflective disc **881** of the rebound check valve **88** is provided with two kidney-shaped openings **871** substantially at the radius of the radially proximal flow passages **872**. In other embodiments of the inventions the deflective discs **873**, **881** could be axially floating and biased by at least one spring.

(29) The coil spring **84** has an end proximal to the piston assembly **4** press fitted on a shoulder nut **43** of the piston assembly **4** and an end distal to the piston assembly **4** press fitted on a tubular projection **833** defined at the compression end of the sleeve **83**.

(30) The hydraulic compression stop assembly **8** includes the sleeve **83** being displaceable along with the piston assembly **4** configured to be slidably introduced inside the inner chamber **811** of the insert **81** at an end of a compression stroke of the hydraulic damper **1** to generate an additional damping force. The functionality of the hydraulic compression stop assembly **8** is illustrated in FIGS. **3-8**.

(31) As shown in FIG. **3** in an inactive state of the hydraulic compression stop assembly **8** during the compression stroke of the hydraulic damper **1** working liquid may flow freely from the compression chamber **12** to the compensation chamber **13**, as illustrated by arrows, both through the first external flow channel **85**, the internal flow channel **86** and through the second external flow channel **89**. During the compression stroke, the deflective disc **881** of the rebound check valve **88** closes radially distal flow passages **882**.

(32) As shown in FIG. **4**, at certain position of the stroke the sleeve **83** enters the insert **81** and its radially external surface begins closing the axially-spaced holes **814** in the insert **81**, which increases flow restrictions, and thereby increases damping. In this embodiment, the first external flow channel **85** remains open, as the flow of liquid is possible through the annular gaps between the axial projections **816**.

(33) As shown in FIG. **5**, at certain further position of the stroke all the axially-spaced holes **814** in the insert **81** are closed by the radially external surface of the sleeve **83** and the first external flow channel **85** is closed. In this position pressure of working liquid in the inner chamber **811** acting on the deflective discs **873** of the compression safety valve **87** in the fixing member **82** reaches a predefined threshold and forces the deflective discs **873** to deflect allowing a flow of working liquid to the second compression valve assembly **72** through the kidney-shaped openings **871** and the radially proximal flow passages **872**.

(34) Further progression of the main piston assembly **4** towards the compression chamber **12** triggers mechanical action of the compression stop assembly **8**. Higher restrictions are generated by the compressing coil spring **84** and by progressively diminishing flow gaps between the coils of the coil spring **84** until finally, as shown in FIG. **6**, the internal flow channel **86** is entirely closed and the coil spring **84** is fully compressed. This substantially shortens the working length of the compression stop assembly and enables a minimum bearing span **32** having excellent properties during the rebound stroke, as illustrated in FIG. **2**. Furthermore, the length and stiffness of the coil spring **84** are additional tuning parameters of the compression stop assembly **8**. The engagement point of the compression stop assembly **8** can be tuned by the length of the coil spring **84**. Relation of the hydraulic and mechanic action of the compression stop assembly **8** can be tuned by the stiffness of the coil spring **84**. The application of the coil spring **84** may reduce the space occupied by the compression stop assembly **8**, when compared with conventional designs.

(35) As also shown in FIG. **6**, at the beginning of the rebound stroke in a fully compressed position of the compression stop assembly **8** the second external flow channel **89** is closed and in this position the flow of working liquid is possible only through radially distal flow passages **882** and the gap between the fixing member **82** and the deflective disc **881** of the rebound check valve **88**. This prevents cavitation.

(36) As shown in FIG. **7**, as the piston assembly **4** moves into the rebound chamber **11** pressure of working liquid in the inner chamber **811** forces the coil spring **84** to extend which opens the radially internal flow channel **86**. Second external flow channel **89** remains closed. In this position the flow is possible only through the rebound check valve **88**.

(37) As the piston assembly **4** advances even further into the rebound chamber **11**, as shown in FIG. **8**, and the fluid flows from the compensation chamber **13** and from the rebound chamber **11** to the compression chamber **12**, through the second external flow channel **89**, the internal flow channel **86**, and the first external flow channel **85**. In this embodiment, the rebound check valve **88** closes when pressure drops below a certain threshold. In other embodiments of the invention the rebound check valve **88** may be configured to remain open during the whole rebound stroke.

(38) Tuning parameters of the compression stop assembly **8** may include number, thickness, diameter and preload of deflective discs **873** of the compression safety valve **87**; number, shape, size, and/or distribution of the axially-spaced holes **814**, the axial grooves **813**, and the axial projections **816**; length of the insert **81** and the sleeve **83**; and finally stiffness and length of the coil spring **84**. Relevant tuning parameters may vary along with the progression of the stroke after the compression stop assembly **8** is activated.

(39) In other embodiments of the invention, not illustrated in detail in the drawing, insert can be made as a single element with the fixing member and provided with the compression safety valve **87** and/or the rebound check valve **88**. In yet other embodiments of the invention, some of the axially-spaced holes **814** may remain open even at the final stage of the compression stroke, so that the application of the compression safety valve **87** and/or the rebound check valve **88** will not be necessary.

(40) The above described embodiments of the present disclosure are merely exemplary. The figures are not necessarily to scale, and some features may be exaggerated or minimized. These and other factors however should not be considered as limiting the spirit of the invention, the intended scope of protection of which is indicated in appended claims.

LIST OF REFERENCE NUMERALS

(41) **1**. damper **11**. rebound chamber **12**. compression chamber **13**. compensation chamber **2**. external tube **3**. main tube **31**. widened section **32**. minimum bearing span **4**. piston assembly **41**. first rebound valve assembly **42**. first compression valve assembly **43**. shoulder nut **5**. piston rod **51**. retaining member **52**. rebound stop **6**. piston rod guide **7**. base valve assembly **71**. second rebound valve assembly **72**. second compression valve assembly **8**. compression stop assembly **81**. insert **811**. inner chamber **812**. annular flange **813**. axial groove **814**. radial hole **815**. annular

channel **816**. axial projection **82**. fixing member **821**. lock washer **822**. dual stroke flow passage **823**. bolt **824**. retainer **83**. sleeve **831**. radial projection **832**. bridge **833**. tubular projection **834**. technological hole **84**. spring **85**. first external flow channel **86**. radially internal flow channel **87**. safety valve **871**. kidney-shaped opening **872**. radially proximal flow passage **873**. deflective disc **88**. rebound check valve **881**. deflective disc **882**. radially distal flow passage **89**. second external flow channel **101**. vehicle chassis **102**. top mount **103**. screw **104**. spring **105**. steering knuckle **106**. vehicle wheel

Claims

1. A hydraulic damper, comprising: a main tube filed with working liquid and extending between an open end and a closed end; a piston assembly slidably disposed inside the main tube, attached to a piston rod that extends outside the hydraulic damper through a sealed piston rod guide located at the open end, dividing the main tube into a rebound chamber and a compression chamber and configured to generate a damping force; a base valve assembly located at the closed end of the compression chamber and configured to control a flow of the working liquid between the compression chamber and a compensation chamber; and a compression stop assembly located in the compression chamber and including an insert defining an inner chamber, and a sleeve displaceable along with the piston assembly and configured to be slidably introduced inside the inner chamber of the insert at an end of a compression stroke of the hydraulic damper to generate an additional damping force; wherein said sleeve has an external diameter that is smaller than an internal diameter of the main tube, defining a first external flow channel between said sleeve and said main tube; wherein said insert defines a second external flow channel extending annularly thereabout between said sleeve and said main tube; and wherein said insert defines a plurality of axially-spaced holes each providing fluid communication between the inner chamber and the base valve assembly via the second external flow channel, wherein said insert adjoins an inner wall of the main tube and said second external flow channel is defined by a plurality of grooves made in a radially external surface of said insert and in fluid connection with said axially-spaced holes.
2. The hydraulic damper according to claim 1, wherein the piston assembly includes compression and rebound valve assemblies to control the flow of working liquid passing between the rebound chamber and the compression chamber to generate the damping force.
3. The hydraulic damper according to claim 1, wherein said sleeve is attached to the piston assembly by a spring disposed within said sleeve.
4. The hydraulic damper according to claim 3, wherein said spring is a coil spring.
5. The hydraulic damper according to claim 1, wherein said insert has an annular flange adjoining an inner wall of the main tube and separating the compression chamber from said second external flow channel.
6. The hydraulic damper according to claim 1, wherein said sleeve has a hollow interior that defines a radially internal flow channel.
7. The hydraulic damper according to claim 1, wherein said sleeve comprises a plurality of radial projections making a sliding fit with the main tube and separated by bridges.
8. The hydraulic damper according to claim 1, wherein the base valve assembly includes a compression valve assembly and a rebound valve assembly each configured to control the flow of the working liquid between the compression chamber and the compensation chamber.
9. The hydraulic damper according to claim 8, wherein the compression stop assembly further comprises a compression safety valve that opens at a predefined threshold of pressure in the inner chamber of said insert and allows for a flow of working liquid from said inner chamber of said insert to the compression valve assembly of the base valve assembly that bypasses said second external flow channel.
10. The hydraulic damper according to claim 8, wherein the compression stop assembly further

comprises a rebound check valve that allows a flow of the working liquid from the rebound valve assembly of the base valve assembly to said inner chamber of said insert, bypassing said second external flow channel during a rebound stroke, and blocking the flow of the working liquid therethrough during the compression stroke.

11. The hydraulic damper according to claim 8, wherein the compression stop assembly further comprises: a compression safety valve that opens at a predefined threshold of pressure in the inner chamber of said insert and allows for a flow of working liquid from said inner chamber of said insert to the compression valve assembly of the base valve assembly that bypasses said second external flow channel; a rebound check valve that allows a flow of the working liquid from the rebound valve assembly of the base valve assembly to said inner chamber of said insert, bypassing said second external flow channel during a rebound stroke, and blocking the flow of the working liquid therethrough during the compression stroke; and a fixing member fixed to the base valve assembly and to said insert and comprising a plurality of angularly disposed dual stroke flow passages joining the base valve assembly with said second external flow channel, wherein at least one of said compression safety valve and said rebound check valve comprises a plurality of deflective discs covering a plurality of angularly disposed flow passages in said fixing member.

12. The hydraulic damper according to claim 1, wherein said insert includes an internal surface defining a plurality of axial projections, and said plurality of axially-spaced holes each include a radially-internal outlet located in a corresponding one of said plurality of axial projections.

13. The hydraulic damper according to claim 1, wherein the main tube includes a widened section to receive the insert.

14. The hydraulic damper according to claim 1, wherein at least one of said insert and said sleeve is made of plastic.

15. The hydraulic damper according to claim 14, wherein said plastic includes polyamide.

16. The hydraulic damper according to claim 1, wherein an insert is made as a single element with a fixing member and provided with at least one of a compression safety valve and a rebound check valve.

17. A hydraulic damper, comprising: a main tube filed with working liquid and extending between an open end and a closed end; a piston assembly slidably disposed inside the main tube, attached to a piston rod that extends outside the hydraulic damper through a sealed piston rod guide located at the open end, dividing the main tube into a rebound chamber and a compression chamber and configured to generate a damping force; a base valve assembly located at the closed end of the compression chamber and configured to control a flow of the working liquid between the compression chamber and a compensation chamber; and a compression stop assembly located in the compression chamber and including an insert defining an inner chamber, and a sleeve displaceable along with the piston assembly and configured to be slidably introduced inside the inner chamber of the insert at an end of a compression stroke of the hydraulic damper to generate an additional damping force; wherein said sleeve has an external diameter that is smaller than an internal diameter of the main tube, defining a first external flow channel between said sleeve and said main tube; wherein said insert defines a second external flow channel extending annularly thereabout between said sleeve and said main tube; and wherein said insert defines a plurality of axially-spaced holes each providing fluid communication between the inner chamber and the base valve assembly via the second external flow channel, wherein the base valve assembly includes a compression valve assembly and a rebound valve assembly each configured to control the flow of the working liquid between the compression chamber and the compensation chamber, wherein the compression stop assembly further comprises: a compression safety valve that opens at a predefined threshold of pressure in the inner chamber of said insert and allows for a flow of working liquid from said inner chamber of said insert to the compression valve assembly of the base valve assembly that bypasses said second external flow channel; a rebound check valve that allows a flow of the working liquid from the rebound valve assembly of the base valve assembly to

said inner chamber of said insert, bypassing said second external flow channel during a rebound stroke, and blocking the flow of the working liquid therethrough during the compression stroke; and a fixing member fixed to the base valve assembly and to said insert and comprising a plurality of angularly disposed dual stroke flow passages joining the base valve assembly with said second external flow channel, wherein at least one of said compression safety valve and said rebound check valve comprises a plurality of deflective discs covering a plurality of angularly disposed flow passages in said fixing member.

18. A hydraulic damper, comprising: a main tube filed with working liquid and extending between an open end and a closed end; a piston assembly slidably disposed inside the main tube, attached to a piston rod that extends outside the hydraulic damper through a sealed piston rod guide located at the open end, dividing the main tube into a rebound chamber and a compression chamber and configured to generate a damping force; a base valve assembly located at the closed end of the compression chamber and configured to control a flow of the working liquid between the compression chamber and a compensation chamber; and a compression stop assembly located in the compression chamber and including an insert defining an inner chamber, and a sleeve displaceable along with the piston assembly and configured to be slidably introduced inside the inner chamber of the insert at an end of a compression stroke of the hydraulic damper to generate an additional damping force; wherein said sleeve has an external diameter that is smaller than an internal diameter of the main tube, defining a first external flow channel between said sleeve and said main tube; wherein said insert defines a second external flow channel extending annularly thereabout between said sleeve and said main tube; and wherein said insert defines a plurality of axially-spaced holes each providing fluid communication between the inner chamber and the base valve assembly via the second external flow channel, wherein said insert includes an internal surface defining a plurality of axial projections, and said plurality of axially-spaced holes each include a radially-internal outlet located in a corresponding one of said plurality of axial projections.
